

# CE 336: Fluid Mechanics laboratory

## Friction head loss in pipes

### Fall 25      Lecture 3

By

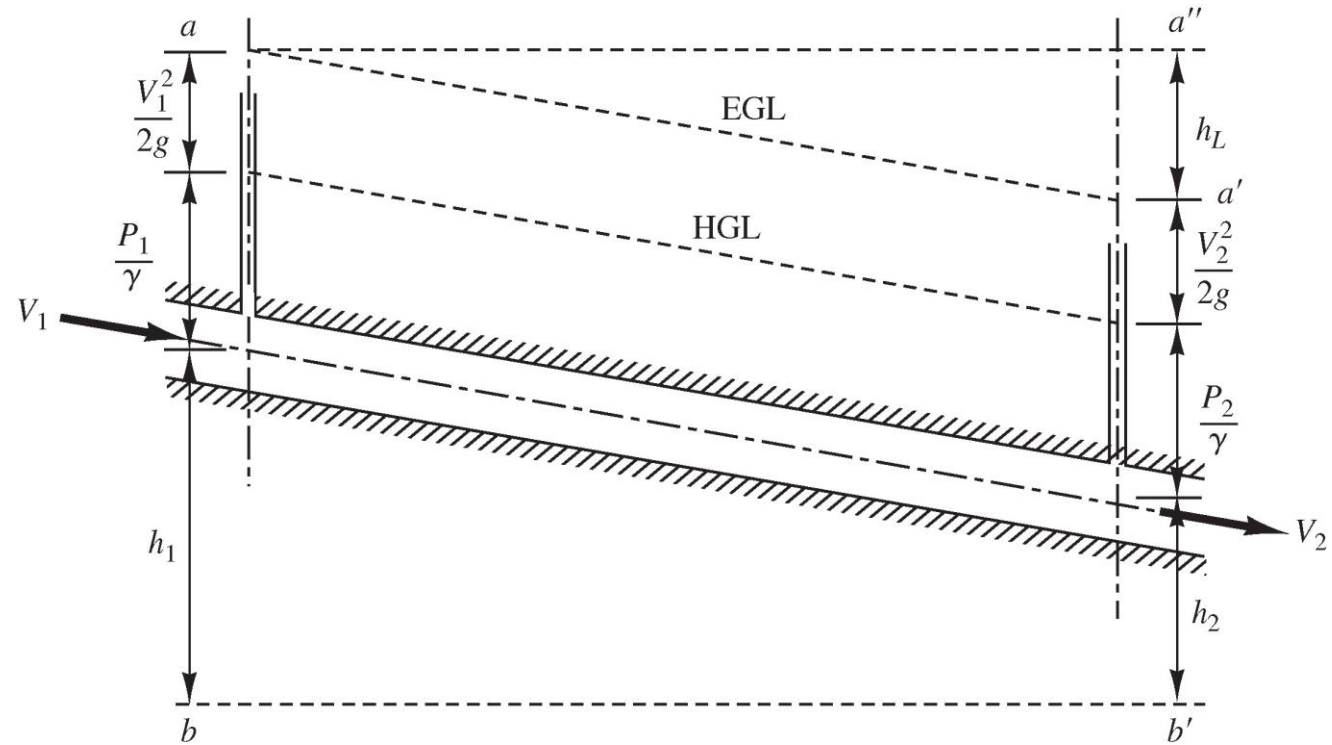
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### 3.5 Loss of Head from Pipe Friction

Energy equation 
$$h_1 + \frac{P_1}{\gamma} + \frac{V_1^2}{2g} = h_2 + \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + h_L$$

■ Total head 
$$H_1 = h_1 + \underbrace{\frac{P_1}{\gamma}}_{\text{HGL}} + \frac{V_1^2}{2g} = \text{EGL}$$



■ Energy loss resulting from friction in a pipeline is commonly termed the *friction head loss* ( $h_f$ ).

- 1) pipe wall friction
- 2) the viscous dissipation in flowing water.

major loss |

and all other losses are referred to as *minor losses*.

Friction head loss :

1. independent of the pressure under which the water flows,
2. linearly proportional to the pipe length ( $L$ ),
3. inversely proportional to some power of the pipe diameter ( $D$ ),
4. proportional to some power of the mean velocity ( $V$ ), and
5. related to the roughness of the pipe, if the flow is turbulent.

The most popular pipe flow equation was derived by Henri Darcy (1803 to 1858), Julius Weisbach (1806 to 1871),

Friction  
head loss :  $h_f = f \left( \frac{L}{D} \right) \frac{V^2}{2g}$

*Darcy–Weisbach equation.*

 = Velocity  
head

$h_f$  and  $V^2/2g$  are both in units of length.

### 3.5.1 Friction Factor for Laminar Flow

$$f = \frac{64\mu}{\rho V D} = \frac{64}{N_R}$$

which indicates a direct relationship between the friction factor ( $f$ ) and the Reynolds number ( $N_R$ ) for laminar pipe flow. It is independent of the surface roughness of the pipe.

### 3.5.2 Friction Factor for Turbulent Flow

When the Reynolds number approaches a higher value—say,  $N_R \gg 2,000$ — **Turbulent flow**

$$h_f = f \left( \frac{L}{D} \right) \frac{V^2}{2g}$$

the value of  $f$  then becomes less dependent on the Reynolds number but

More dependent on relative roughness  $(e/D)$  of the pipe.

■ The quantity  $e$  is a measure of the average roughness height of the pipe wall irregularities  
 $D$  is the pipe diameter.

Table 3.1 Roughness Heights,  $e$ , for Certain Common Pipe Materials

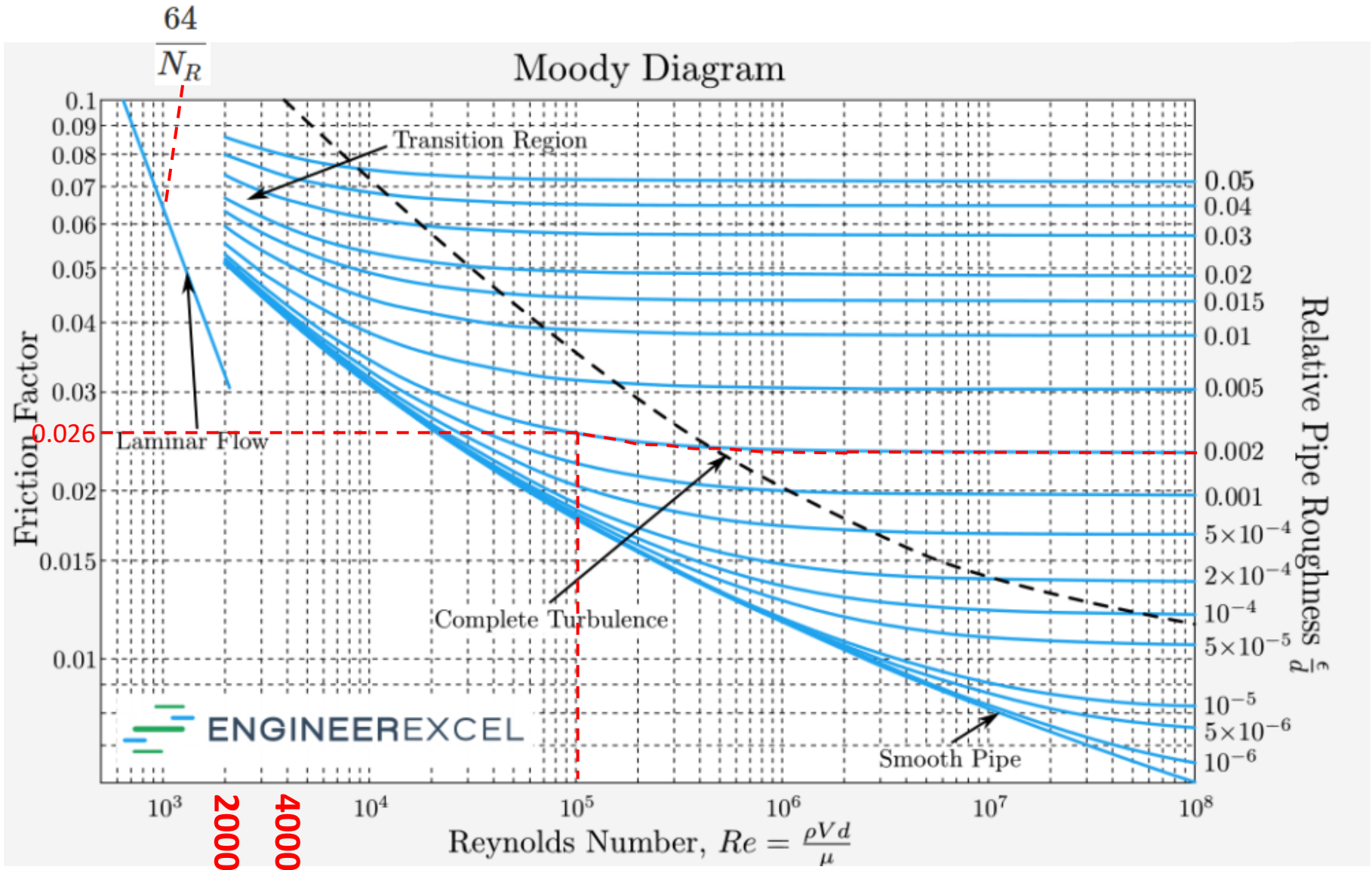
| Pipe Material | $e$ (mm) | $e$ (ft) |
|---------------|----------|----------|
| Brass         | 0.0015   | 0.000005 |

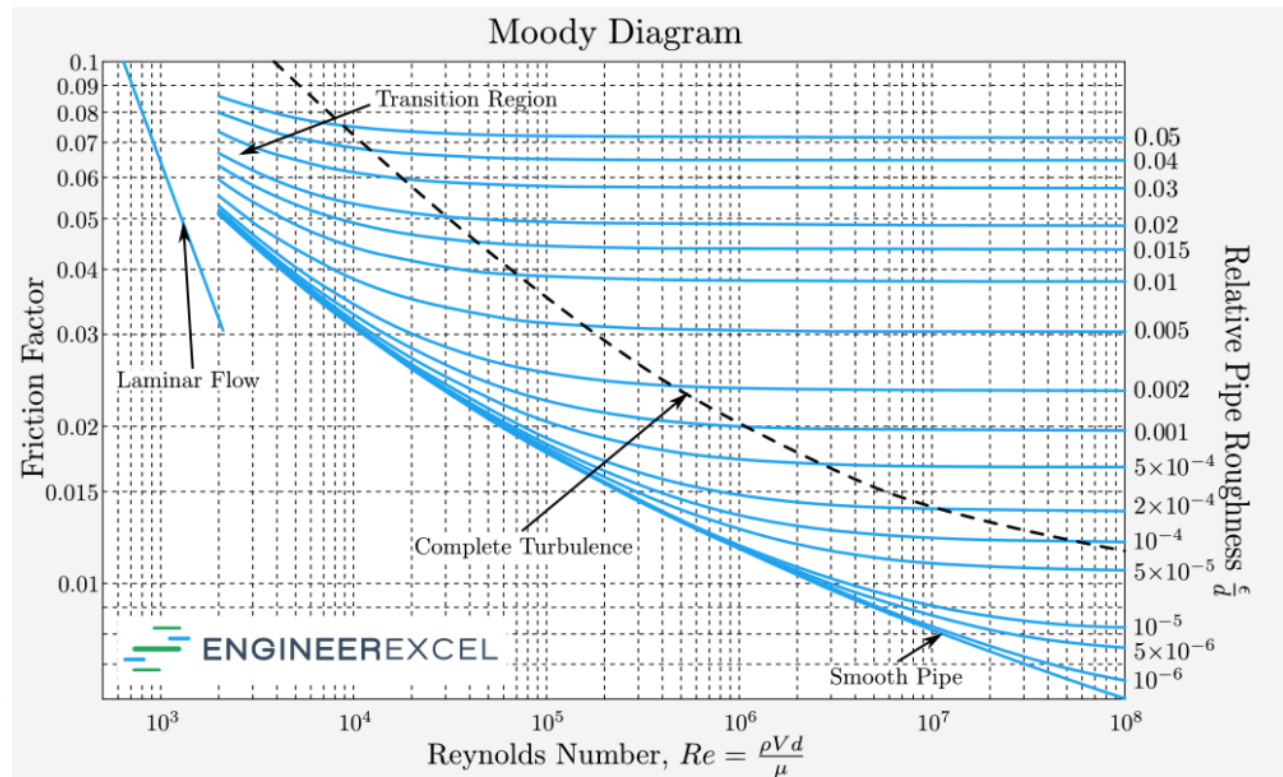
■ von Kármán found that  $f$  can be expressed as

$$\frac{1}{\sqrt{f}} = 2 \log \left( 3.7 \frac{D}{e} \right)$$

$$\frac{1}{\sqrt{f}} = 2 \log \left( 3.7 \frac{D}{e} \right)$$

convenient chart was prepared by Lewis F. Moody† (Figure 3.8); it is commonly called the *Moody diagram of friction factors for pipe flow*.





The chart clearly shows the four zones of pipe flow:

1. a laminar flow zone where the friction factor is a simple linear function of the Reynolds number,
2. a critical zone where values are uncertain because the flow might be neither laminar nor truly turbulent,
3. a transitional zone where  $f$  is a function of both the Reynolds number and the relative roughness of the pipe, and
4. a zone of fully developed turbulence where the value of  $f$  depends solely on the relative roughness and is independent of the Reynolds number.

**Figure 3.8**  may be used together with **Table 3.1**  to obtain the friction factor  $f$  for circular pipes.

## Calculations

1. Calculate the flow rate using the volume of water collected in the hydraulic bench and time to collect the water - Column #3 of the Tables 1 and 2.
2. Calculate flow velocity  $V$  using flow rate,  $Q = VA$  - Column 5 of Tables 1 and 2.
3. Calculate Reynold's number using Equation 1.-Column 6 of Tables 1 and 2.
4. Find the friction factor  $f$  from Moody diagram – Column 7 of Tables 1 and 2.
5. Calculate the friction loss in pipes using Darcy Weisbach equation (Equation 2) –Column 8 of Tables 1 and 2.
6. Plot a graph of measured head loss,  $h_m$  versus velocity  $V$  for each size of pipe. Identify laminar, transitional and turbulent flow on the graphs. Establish that  $h_m$  vs  $V$  graph is a straight line for laminar flow.
7. Plot a graph of  $\log h_m$  versus  $\log V$  for each size of pipes. Establish that the graph is a straight line for the turbulent flow ( $h_m \propto V^n$ ). Determine the slope of the straight line to find  $n$ .
8. Estimate the value of Reynolds number ( $Re$ ) at the start and finish of the transitional flow. The two velocities corresponding to these two values of  $Re$  are called upper and lower critical velocities.

## **Discussions**

Discuss your results by addressing the followings-

1. Compare the head loss values computed using Darcy Weisbach's equation and with the corresponding measured values. Discuss if the Darcy Weisbach's equation can be applied to calculate friction losses if fluid velocity and pipe dimensions are known.
2. Discuss the Reynold's number observed for laminar flow and turbulent flow.
3. Discuss if the head losses increased with the increase in Reynold's number. Which pipe size has maximum head loss?



